

Questions with Answer Keys

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Q1 - Relation between roots and coefficients

The real roots of the equation $x^2 + 5|x| + 4 = 0$ are

- (1) $\{-1, -4\}$
- (2) $\{1, 4\}$
- (3) $\{-4, 4\}$
- (4) None of these

Q2 - Relation between roots and coefficients

The value of a for which the sum of the squares of the roots of the equation $x^2 - (a-2)x - a - 1 = 0$ assumes the least value is

- (1) 2
- (2) 3
- (3) 0
- (4) 1

Q3 - Relation between roots and coefficients

Let α and β be the roots of equation $x^2 - 6x - 2 = 0$ with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is

- (1) 1
- (2) 2
- (3) 3
- (4) None of these

Q4 - Relation between roots and coefficients

Let k be a real number such that $k \neq 0$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -2k$ and $\alpha^2 + \beta^2 = 4k^2 - 2k$, then a quadratic equation having $\frac{\alpha+\beta}{\alpha}$ and $\frac{\alpha+\beta}{\beta}$ as its roots, equals

- (1) $4x^2 - 4kx + k = 0$

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Questions with Answer Keys

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(2) $x^2 - 4kx + 4k = 0$

(3) $4kx^2 - 4x + k = 0$

(4) $4kx^2 - 4kx + 1 = 0$

Q5 - common root

If $(x + k)$ is a common factor of the expressions $x^2 + px + q$ and $x^2 + lx + m$, then k is equal to

(1) $\frac{p+q}{l+m}$

(2) $\frac{p-l}{q-m}$

(3) $\frac{q+m}{p+l}$

(4) $\frac{q-m}{p-l}$

Q6 - common root

If the quadratic equations $3x^2 + ax + 1 = 0$ and $2x^2 + bx + 1 = 0$ have a common real root, then the value of the expression $5ab - 2a^2 - 3b^2 = 0$ is

(1) 0

(2) 1

(3) -1

(4) None of these

Q7 - common root

If the equations $ax^2 + bx + c = 0$ and $x^3 + 3x^2 + 3x + 2 = 0$ have two common roots, then

(1) $a = 2b = c$

(2) $a = b = c$

(3) $b^2 = 4ac$

(4) None of these

Q8 - common root

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Questions with Answer Keys

#MathBoleTohMathonGo

If the quadratic equations $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ ($a \neq 0$) have a common root, then the numerical value of $(a + b)$ is

- (1) 0
- (2) 1
- (3) -1
- (4) 2

Q9 - graphs for quadratic

If $ax^2 + bx + c = 0$ does not have real and distinct roots then

- (1) $a(a + b + c) > 0$
- (2) $(a + b + c) > 0$
- (3) $b(a + b + c) > 0$
- (4) None of these

Q10 - graphs for quadratic

If $ax^2 + bx + 10 = 0$ does not have real and distinct roots, then the minimum value of $(5a - b)$ is

- (1) -3
- (2) -2
- (3) -1
- (4) 0

Q11 - graphs for quadratic

While applying the quadratic formula to a quadratic equation $f(x) = 0$ where

$f(x) = ax^2 + bx + c$ ($a, b, c \in R$ and $a \neq 0$), it happens that $c = \frac{b^2}{4a}$. Then, the graph of $y = f(x)$ will certainly

- (1) have a maximum
- (2) have a minimum

Questions with Answer Keys

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(3) be a tangent to the X -axis(4) lie completely above X -axis

Q12 - graphs for quadratic

The number of integral values of α for which the quadratic expression $\alpha x^2 + |2\alpha - 3|x - 6$ is positive for exactly two integral values of x is equal to

(1) 3

(2) 2

(3) 1

(4) 0

Q13 - graphs for quadratic

If $\min(2x^2 - ax + 2) > \max(b - 1 + 2x - x^2)$. Then roots of the equation $2x^2 + ax + (2 - b) = 0$ are

(1) positive and distinct

(2) negative and distinct

(3) opposite in sign

(4) imaginary

Q14 - Range of quadratic expressions

If x is real, then the maximum value of $\frac{3x^2 + 9x + 17}{3x^2 + 9x + 7}$ is

(1) 41

(2) 1

(3) $\frac{17}{7}$ (4) $\frac{1}{4}$

Q15 - Range of quadratic expressions

If x is real, then the value of the expression $\frac{(x+m)^2 - 4mn}{2(x-n)}$ will not

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(1) lie between m and $m + n$ (2) lie between $2m$ and $2n$ (3) greater than $m + 2n$ (4) greater than $m + n$

Q16 - Location of roots

If both the roots of the quadratic equation $x^2 - 2kx + (k^2 + k - 5) = 0$ are less than 5, then k lies in the interval

(1) (4,5)

(2) $(-\infty, 4)$ (3) $(6, \infty)$

(4) (5,6)

Q17 - Location of roots

All the values of m for which both the roots of $x^2 - 2mx + m^2 - 1 = 0$ are greater than -2 but less than 4, lie in the interval

(1) $m > 3$ (2) $-1 < m < 3$ (3) $1 < m < 4$ (4) $-2 < m < 0$

Q18 - Location of roots

If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then

(1) $a < 2$ (2) $2 \leq a \leq 3$ (3) $3 < a \leq 4$ (4) $a > 4$

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Questions with Answer Keys

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Q19 - Location of roots

If exactly one root of the quadratic equation $x^2 - \left(k + \frac{11}{3}\right)x - (k^2 - k + 1) = 0$ lies in $(0,3)$, then which one of the following relation is correct?

(1) $-3 < k < -1$

(2) $-8 < k < -4$

(3) $1 < k < 4$

(4) None of these

Q20 - Location of roots

Number of integral values of k for which exactly one root of the equation $5x^2 + (k+1)x + k = 0$ lies in the interval $(1,3)$, is

(1) 5

(2) 6

(3) 7

(4) 8

Questions with Answer Keys

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Answer Key

Q1 (4)	Q2 (4)	Q3 (3)	Q4 (2)
Q5 (4)	Q6 (2)	Q7 (2)	Q8 (3)
Q9 (1)	Q10 (2)	Q11 (3)	Q12 (4)
Q13 (4)	Q14 (1)	Q15 (2)	Q16 (2)
Q17 (2)	Q18 (1)	Q19 (1)	Q20 (4)